TMUA / WORKED SOLUTIONS

Set A Paper 1

Applications of Mathematical Knowledge 20 questions

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Exponentials & Logarithms

1. Find the complete set of positive real values of x, with $x \neq 1$, for which

$$\log_2 x < \log_x 2$$

A $0 < x < \frac{1}{2}$ or $1 < x < 2$

B $\frac{1}{2} < x < 1$ or $x > 2$

C $1 < x < 2$

D $0 < x < \frac{1}{2}$

E $0 < x < 1$ or $x > 2$

F $x > 2$

G $0 < x < 2$, $x \neq 1$

Solution **Answer A**

Let

$$y = \log_2 x$$

Then $x = 2^y$. Since $x \neq 1$, we have $y \neq 0$. Also, $x = 2^y$ gives $2 = x^{1/y}$, so

$$\log_x 2 = \frac{1}{y}$$

The inequality therefore becomes

$$y < \frac{1}{y}$$

We must consider the sign of y before multiplying by it.

If $y > 0$, multiplication by y preserves the inequality:

$$y^2 < 1$$

Together with $y > 0$, this gives $0 < y < 1$. As $x = 2^y$, this corresponds to

$$1 < x < 2$$

If $y < 0$, multiplication by y reverses the inequality:

$$y^2 > 1$$

Together with $y < 0$, this gives $y < -1$. Hence

$$0 < x < \frac{1}{2}$$

The reversal in the second case is essential; multiplying by a negative value without reversing the inequality would incorrectly produce $\frac{1}{2} < x < 1$. Thus the complete set is $0 < x < \frac{1}{2}$ or $1 < x < 2$, so the correct option is **A**.

Remark:

Note that we can skip the initial rearrangement of $y = \log_2 x$ into $\log_x 2 = \dfrac{1}{y}$ if we know the entertaining identity

$$\log_x 2 = \frac{1}{\log_2 x}$$

which, with $y = \log_2 x$, immediately turns the inequality into

$$y < \frac{1}{y}$$

This identity is a specific case of the rule

$$\log_b a = \frac{1}{\log_a b}$$

which itself follows from the general change-of-base formula for logarithms:

$$\log_b a = \frac{\log_c a}{\log_c b}$$

Knowledge of the change-of-base formula is not on the current TMUA specification, but that does not mean it isn't worth knowing!

Inequalities

2. Fix a real number $a \neq 0$.

Consider the inequality

$$|x - a| + |x + a| < 3|a|$$

Which of the following describes the set of all real x satisfying the inequality?

A No real solutions

B All real x with $|x| < |a|$

C All real x with $|x| < \dfrac{3|a|}{2}$

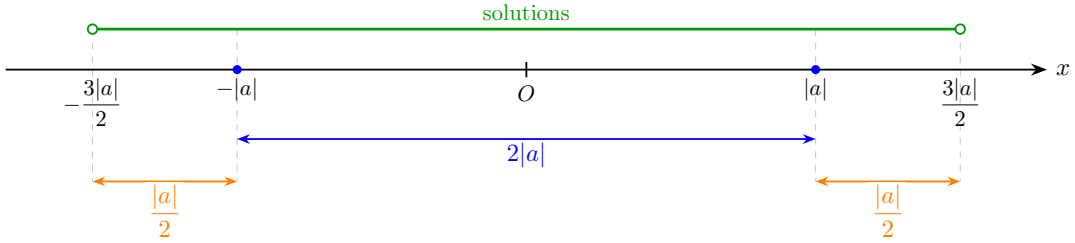
D All real x with $|x| < 3|a|$

E All real x with $\dfrac{|a|}{2} < |x| < \dfrac{3|a|}{2}$

Solution **Answer C**

The two points a and $-a$ are (in some order) the points $-|a|$ and $|a|$ on the number line.

Each term on the left-hand side is a distance: $|x - a|$ is the distance from x to a, and $|x + a|$ is the distance from x to $-a$.



If

$$-|a| \leq x \leq |a|$$

the two distances together equal the distance between the two fixed points, which is

$$2|a|$$

Hence every such x satisfies the inequality because

$$2|a| < 3|a|$$

Now suppose x lies outside this central interval, at a distance d from its nearer endpoint. Its distances from the two fixed points are then d and $2|a| + d$, so their sum is

$$2|a| + 2d$$

We require

$$2|a| + 2d < 3|a|$$

and hence

$$d < \frac{|a|}{2}$$

Thus there is an additional distance of $\frac{|a|}{2}$ available beyond each end of the central interval. The complete range is therefore

$$-\frac{3|a|}{2} < x < \frac{3|a|}{2}$$

The endpoints are excluded because they make the original left-hand side equal to $3|a|$, rather than less than $3|a|$

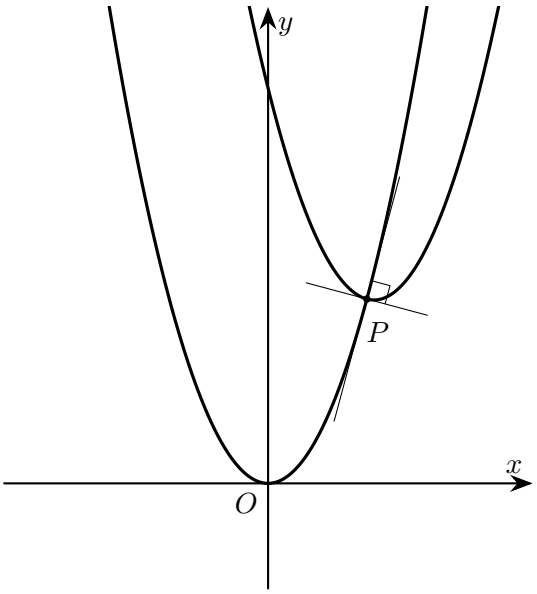
This is equivalent to

$$|x| < \frac{3|a|}{2}$$

Therefore the correct option is **C**.

Differentiation

3. The diagram shows parts of the curves $y = x^2$ and $y = (x-2)^2 + b$, where b is positive.
The curves meet at P, and the tangents to the curves at P are perpendicular.
What is the value of b?



A $\frac{\sqrt{3}}{2}$

B $\sqrt{3}$

C $2\sqrt{3}$

D $3\sqrt{3}$

E $4\sqrt{3}$

Solution **Answer C**

Let x be the x-coordinate of P.

Since P lies on both curves, their y-coordinates are equal:

$$x^2 = (x-2)^2 + b$$

Expanding and rearranging gives

$$\begin{aligned} b &= 4x - 4 \\ &= 4(x-1) \end{aligned}$$

For the second curve, first write

$$y = x^2 - 4x + 4 + b$$

For the first curve, $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x$ and for the second curve $\frac{\mathrm{d}y}{\mathrm{d}x} = 2x - 4$

The gradients of the two curves at P are therefore $2x$ and $2x-4$.

Perpendicular lines have gradients that are negative reciprocals, so the product of their gradients is -1:

$$(2x)(2x-4) = -1$$

Hence

$$4x^2 - 8x + 1 = 0$$

and the quadratic formula gives

$$x = \frac{8 \pm \sqrt{64-16}}{8} = 1 \pm \frac{\sqrt{3}}{2}$$

For

$$x = 1 + \frac{\sqrt{3}}{2},$$

we obtain

$$b = 4(x-1) = 4\left(\frac{\sqrt{3}}{2}\right) = 2\sqrt{3} > 0$$

For

$$x = 1 - \frac{\sqrt{3}}{2}$$

we instead obtain

$$b = 4(x-1) = 4\left(-\frac{\sqrt{3}}{2}\right) = -2\sqrt{3} < 0$$

which must be rejected since b is positive.

Therefore

$$b = 2\sqrt{3}$$

and the correct option is **C**.

Algebra

4. A positive real number x satisfies

$$(x-2)(x+3)(x+4)(x+9) = 3024$$

What is the value of $x^2 + 7x$?

A 16

B 38

C 54

D 60

E 114

Solution **Answer D**

Expanding all four factors would produce a quartic and create unnecessary work.

Instead, notice that the two pairs $(x+3)(x+4)$ and $(x-2)(x+9)$ both have $7x$ as their linear term.

Set

$$u = x^2 + 7x$$

Then

$$\begin{aligned}(x+3)(x+4) &= x^2 + 7x + 12 \\ &= u + 12\end{aligned}$$

and

$$\begin{aligned}(x-2)(x+9) &= x^2 + 7x - 18 \\ &= u - 18\end{aligned}$$

so the equation becomes

$$(u-18)(u+12) = 3024$$

At this point, testing the options can be faster. For $u = 60$,

$$(u-18)(u+12) = 42 \times 72 = 3024$$

Thus the answer is option **D.**

For a complete algebraic solution, expanding gives

$$u^2 - 6u - 3240 = 0$$

To factorise this, we need two numbers with product -3240 and sum -6. Since their sum is small, their magnitudes should be close. The prime factorisation helps to find them:

$$3240 = 2^3 \times 3^4 \times 5 = (2 \times 3^3)(2^2 \times 3 \times 5) = 54 \times 60$$

The required numbers are therefore 54 and -60, giving

$$(u - 60)(u + 54) = 0$$

Thus $u = 60$ or $u = -54$.

Since x is positive, $u = x(x + 7) > 0$, so $u = -54$ is impossible. Hence $x^2 + 7x = 60$, which is option **D**.

This question depends heavily on spotting the useful pairing of factors. If that pairing is not apparent quickly under exam conditions, it is sensible to skip the question and return later rather than expand a quartic.

Integration

5. The curve $y = x^3 - 3x + 5$ has a tangent line at the point where $x = 2$.

This tangent meets the curve again at exactly one other point.

Find the area of the region enclosed between the curve and the tangent.

A -108

B 54

C 96

D 108

E 120

F 216

Solution **Answer D**

Differentiate the equation of the curve:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 3$$

At $x = 2$, the gradient is $3(2^2) - 3 = 9$, and the point on the curve is $(2, 7)$. Hence the tangent has equation

$$y - 7 = 9(x - 2)$$

so

$$y = 9x - 11$$

To find the other intersection, equate the curve and tangent:

$$x^3 - 3x + 5 = 9x - 11$$

Therefore

$$x^3 - 12x + 16 = 0$$

Since the tangent touches the curve at $x = 2$, the factor $(x - 2)$ of this cubic occurs twice as $(x - 2)^2$. The other factor must therefore be $(x + 4)$ to make the leading term of x^3 and the constant term of 16.

Hence, or otherwise, this cubic factorises as

$$x^3 - 12x + 16 = (x - 2)^2(x + 4)$$

Thus the other intersection is at $x = -4$.

The area between curves $y = f(x)$ and $y = g(x)$ from $x = a$ to $x = b$, when $f(x) \geq g(x)$, is

$$A = \int_a^b \big(f(x) - g(x)\big)\,\mathrm{d}x$$

Here, one can check that the curve lies above the tangent for $-4 \leq x \leq 2$ so this formula is appropriate. Hence

$$
\begin{aligned}
A &= \int_{-4}^{2} (x^3 - 3x + 5) - (9x - 11)\,\mathrm{d}x \\
&= \int_{-4}^{2} (x^3 - 12x + 16)\,\mathrm{d}x \\
&= \left[\frac{x^4}{4} - 6x^2 + 16x\right]_{-4}^{2} \\
&= 12 - (-96) \\
&= 108
\end{aligned}
$$

Therefore the correct option is **D**.

Coordinate Geometry

6. A straight line passes through the point $(2, 5)$ and has gradient m.

The line forms a triangle of area 25 with the positive x-axis and the positive y-axis.

What is the sum of all possible values of m?

A -15

B $-\frac{15}{2}$

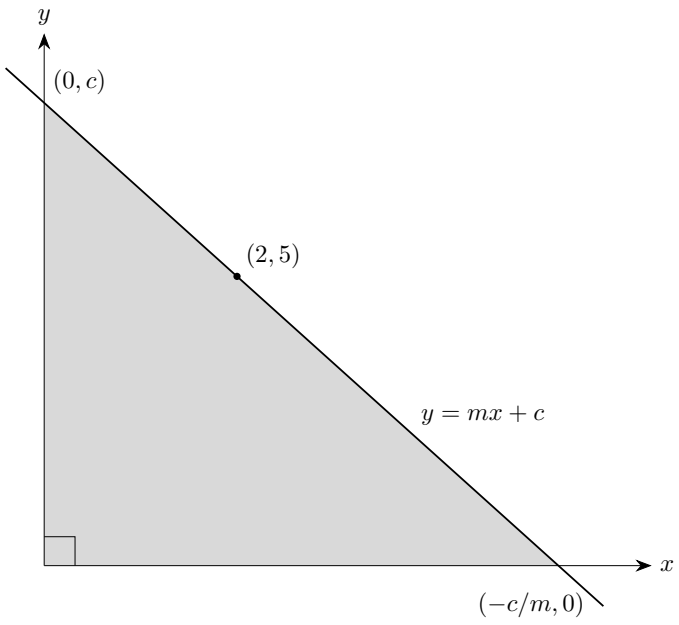
C $-\frac{25}{4}$

D $-\frac{15}{4}$

E $\frac{15}{2}$

Solution **Answer B**

The geometry is represented by the following sketch.



The two intercepts must both lie on the positive axes, so the line slopes downwards from left to right. Hence $m < 0$.

Write the equation of the line as

$$y = mx + c$$

Since it passes through $(2, 5)$,

$$5 = 2m + c$$

so its y-intercept is

$$c = 5 - 2m$$

At the x-intercept, $y = 0$, so

$$0 = mx + c$$

and therefore the x-intercept is $-\dfrac{c}{m}$. This is positive because $c > 0$ and $m < 0$.

The triangle consequently has base $-\dfrac{c}{m}$ and height c. Using its area gives

$$25 = \frac{1}{2}\left(-\frac{c}{m}\right)c = -\frac{c^2}{2m}$$

Substituting $c = 5 - 2m$ and multiplying by $2m$, which is non-zero, gives

$$(5 - 2m)^2 = -50m$$

Therefore

$$4m^2 + 30m + 25 = 0$$

The quadratic formula gives

$$m = \frac{-30 \pm \sqrt{30^2 - 4 \times 4 \times 25}}{8} = \frac{-15 \pm 5\sqrt{5}}{4}$$

Both values are negative and hence both give positive intercepts, so both are valid. Their sum is

$$\frac{-15 + 5\sqrt{5}}{4} + \frac{-15 - 5\sqrt{5}}{4} = -\frac{15}{2}$$

As a shortcut, for $Ax^2 + Bx + C = 0$, the sum of the roots is $-\dfrac{B}{A}$ and their product is $\dfrac{C}{A}$.

Thus the sum could have been read directly from $4m^2 + 30m + 25 = 0$ as $-\dfrac{30}{4} = -\dfrac{15}{2}$.

Therefore the correct option is **B**.

Number

7. Consider the integer

$$N = 8^{15} \times 5^{40}$$

When N is written out in full, how many digits does it have?

A 40

B 41

C 42

D 43

E 46

F 55

Solution **Answer C**

Calculating N in full is inefficient.

Instead, pair factors of 2 and 5 to produce powers of 10.

Since $8^{15} = 2^{45}$,

$$\begin{aligned} N &= 2^{45} \times 5^{40} \\ &= 2^5 \times 2^{40} \times 5^{40} \\ &= 32 \times (2 \times 5)^{40} \\ &= 32 \times 10^{40} \\ &= 3.2 \times 10^{41} \end{aligned}$$

Thus $10^{41} \leq N < 10^{42}$, so N has 42 digits.

Equivalently, 32×10^{40} is 32 followed by 40 zeros, giving $2 + 40 = 42$ digits.

Therefore the correct option is **C**.

Sequences & Series

8. The nth term of a sequence is a quadratic expression in n.
The second, fifth and eighth terms are 4, 13 and 40, respectively.
What is the eleventh term?

A 67

B 76

C 85

D 94

E 121

Solution **Answer C**

Write the nth term as

$$an^2 + bn + c$$

Using the three given terms gives

$$4a + 2b + c = 4$$
$$25a + 5b + c = 13$$
$$64a + 8b + c = 40$$

Subtracting the first equation from the second, and the second from the third, eliminates c:

$$21a + 3b = 9$$
$$39a + 3b = 27$$

Subtracting these equations gives $18a = 18$, so $a = 1$.
Hence

$$21 + 3b = 9$$

giving $b = -4$.
Substitution into the first original equation then gives

$$4 - 8 + c = 4$$

so $c = 8$. Therefore the nth term is

$$n^2 - 4n + 8$$

so, in particular, the 11th term is

$$11^2 - 4(11) + 8 = 121 - 44 + 8 = 85$$

Thus the correct option is **C**.

Alternative Solution:

Recall that a quadratic sequence has constant second differences.

Here we are looking only at the terms in positions

$$2,\ 5,\ 8,\ 11,\ \ldots$$

These positions are equally spaced: each time, the term number increases by 3. For any quadratic sequence, moving forward by the same fixed number of places each time produces another quadratic sequence.

In fact, if the original second difference is d, then moving three places at a time gives second difference

$$3^2 d = 9d$$

So the terms in positions $2, 5, 8, 11, \ldots$ can themselves be treated as a quadratic sequence.

The first three of these terms are

$$4,\ 13,\ 40$$

Their first differences are

$$13 - 4 = 9, \qquad 40 - 13 = 27$$

so the second difference is

$$27 - 9 = 18$$

The next first difference must therefore be

$$27 + 18 = 45$$

Hence the next term is

$$40 + 45 = 85$$

so the correct option is again **C**.

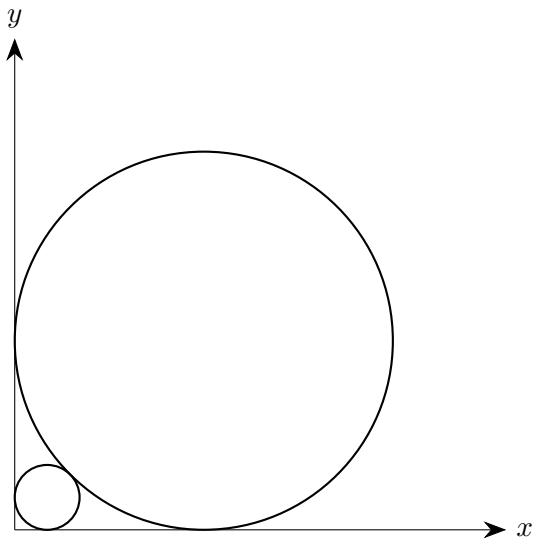
Geometry

9. A circle of radius 1 lies in the region where $x \geq 0$ and $y \geq 0$, and touches both axes.

A second, smaller circle also lies in this region, touches both axes, and touches the first circle at exactly one point.

The two circles do not overlap.

What is the radius of the smaller circle?



A $\frac{1}{3}$

B $3 - 2\sqrt{2}$

C $\sqrt{2} - 1$

D $\frac{\sqrt{2} - 1}{2}$

E $2 - \sqrt{2}$

F $17 - 12\sqrt{2}$

G $3 + 2\sqrt{2}$

Solution **Answer B**

Let the smaller radius be r.

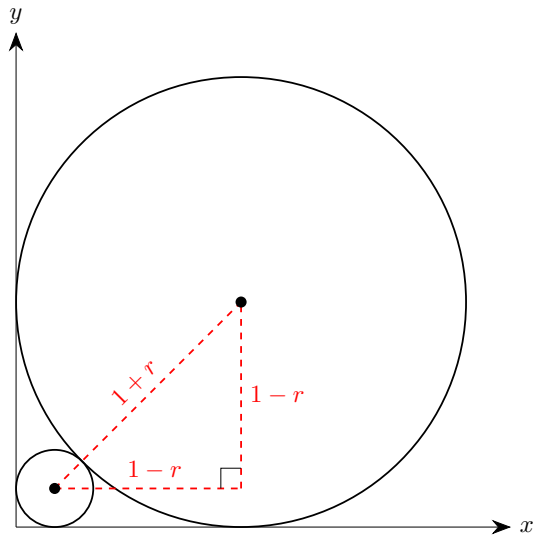
A circle tangent to both axes has its centre one radius from each axis.

Therefore the centres of the large and small circles are $(1, 1)$ and (r, r) respectively.

The horizontal and vertical distances between the centres are both $1 - r$.

Also, because the circles touch externally without overlapping, the distance between their centres is the sum of their radii, $1 + r$.

See the following diagram for an illustration:



The line joining the centres is therefore the hypotenuse of a right-angled triangle whose other two sides both have length $1 - r$.

Pythagoras' theorem gives

$$(1 + r)^2 = (1 - r)^2 + (1 - r)^2$$

Expanding and rearranging gives

$$1 + 2r + r^2 = 2 - 4r + 2r^2$$

$$r^2 - 6r + 1 = 0$$

Using the quadratic formula,

$$r = \frac{6 \pm \sqrt{36 - 4}}{2} = 3 \pm 2\sqrt{2}$$

The smaller circle must have $0 < r < 1$. The root $3 + 2\sqrt{2}$ is greater than 1, so it is rejected.

Also, $2\sqrt{2} < 3$ and $2\sqrt{2} > 2$, so $0 < 3 - 2\sqrt{2} < 1$.

Therefore the radius is $3 - 2\sqrt{2}$, which is option **B**.

Sequences & Series

10. A sequence a_n is defined by $a_0 = 3$ and, for $n \geq 0$,

$$a_{n+1} = 3(a_n)^2$$

Which of the following is equal to $\log_3(a_n)$?

A 2^n

B $2^n - 1$

C $2^{n+1} - 1$

D $2^{n+1} + 1$

E $2^n - 2$

F $2^{n+1} - 2$

Solution **Answer C**

Trying the options is exceptionally effective here. Since $a_0 = 3$,

$$\log_3(a_0) = 1$$

At $n = 0$, option A gives $2^0 = 1$ and option C gives $2^1 - 1 = 1$. Every other option gives a value different from 1, so only A and C remain.

Now

$$a_1 = 3(3)^2 = 27$$

Therefore $\log_3(a_1) = 3$. At $n = 1$, option A gives 2, whereas option C gives $2^2 - 1 = 3$. Hence the correct option is **C**.

Alternative Solution:

Now let's actually solve the problem by finding the nth term. Let

$$b_n = \log_3(a_n)$$

Taking logarithms of the recurrence gives

$$\begin{aligned} b_{n+1} &= \log_3 a_{n+1} \\ &= \log_3\left(3(a_n)^2\right) \\ &= 1 + 2\log_3 a_n \\ &= 1 + 2b_n \end{aligned}$$

This recurrence contains both multiplication and addition. The sequence is therefore neither arithmetic or geometric, but a mixture of the two.

However, we can turn it into a geometric sequence. Adding 1 to both sides:

$$b_{n+1} + 1 = 2(b_n + 1)$$

Define $c_n = b_n + 1$. Then

$$c_{n+1} = 2c_n$$

So c_n is a geometric sequence with common ratio 2. Also, $b_0 = \log_3 3 = 1$, so $c_0 = 2$. Thus the first term of the geometric sequence is 2.

The usual geometric sequence formula for the nth term is ar^{n-1} for the term in position n.

Here c_n is in position $n + 1$ because the indexing begins at $n = 0$.

Therefore

$$\begin{aligned} c_n &= ar^{(n+1)-1} \\ &= 2 \cdot 2^{(n+1)-1} \\ &= 2^{n+1} \end{aligned}$$

Since $b_n = c_n - 1$,

$$\log_3(a_n) = b_n = 2^{n+1} - 1$$

Which again obtains option **C**.

The full recurrence method is instructive but far more involved, making this a good example of when trying the options first is the right exam technique.

Number

11. An integer n is chosen at random from the integers 1 to 1260 inclusive, with each integer equally likely to be chosen.

The fraction $\frac{n}{1260}$ is written as a decimal.

What is the probability that this decimal terminates and that its shortest form has exactly two digits after the decimal point?

In the shortest form, zeros at the end are omitted; for example, 0.50 is written as 0.5.

A $\frac{1}{630}$

B $\frac{2}{315}$

C $\frac{1}{126}$

D $\frac{1}{90}$

E $\frac{1}{70}$

F $\frac{1}{63}$

Solution **Answer C**

First recall when a fraction has a terminating decimal.

A fraction in its lowest terms has a terminating decimal if and only if its denominator has no prime factors other than 2 and 5.

The reason is that a terminating decimal can be written using a denominator which is a power of 10, and

$$10^r = 2^r 5^r$$

For example,

$$\frac{3}{8} = \frac{375}{1000} = 0.375$$

because the denominator $8 = 2^3$ can be turned into a power of 10 by multiplying by a suitable power of 5.

Now consider the number of digits after the decimal point.

A fraction in lowest terms has at most two digits after the decimal point exactly when its denominator divides

$$100 = 2^2 \times 5^2$$

Similarly, it has at most one digit after the decimal point exactly when its denominator divides

$$10 = 2 \times 5$$

Therefore, for the decimal to have **exactly two digits** in its shortest form, its reduced denominator must divide 100 but must not divide 10.

Now factorise the denominator in the question:

$$1260 = 2^2 \times 3^2 \times 5 \times 7$$

After reducing

$$\frac{n}{1260}$$

its denominator must still be a divisor of 1260.

So the reduced denominator must divide both 1260 and 100. Their common prime factors give

$$\text{hcf}(1260, 100) = 2^2 \times 5 = 20$$

Hence the possible reduced denominators are the divisors of 20:

$$1,\ 2,\ 4,\ 5,\ 10,\ 20$$

But

$$1,\ 2,\ 5,\ 10$$

all divide 10, so these would give at most one digit after the decimal point.

Thus the only possible reduced denominators are

$$4 \qquad \text{or} \qquad 20$$

We now count how many values of n give each denominator.

For reduced denominator 4, write

$$\frac{n}{1260} = \frac{a}{4}$$

where $\frac{a}{4}$ is in lowest terms.

Since

$$1260 = 4 \times 315$$

we have

$$n = 315a$$

To keep the reduced denominator equal to 4, the numerator a must have no common factor with 4. Among

$$1, 2, 3, 4$$

the possible values are therefore

$$a = 1,\ 3$$

So there are 2 successful values of n with reduced denominator 4.

For reduced denominator 20, write

$$\frac{n}{1260} = \frac{a}{20}$$

in lowest terms.

Since

$$1260 = 20 \times 63$$

we have

$$n = 63a$$

Now a must have no common factor with 20. Since

$$20 = 2^2 \times 5$$

this means that a must be divisible by neither 2 nor 5.

Among the integers from 1 to 20, these are

$$1,\ 3,\ 7,\ 9,\ 11,\ 13,\ 17,\ 19$$

So there are 8 successful values of n with reduced denominator 20.

Hence the total number of successful choices is

$$2 + 8 = 10$$

There are 1260 equally likely possible values of n, so the required probability is

$$\frac{10}{1260} = \frac{1}{126}$$

Therefore the correct option is **C**.

Functions & Graphs

12. A function f has the form $f(x) = ax + b$, where a and b are real constants.
For every real x, applying f twice gives

$$f(f(x)) = 4x + 3$$

What is the difference between the greatest and least possible values of $f(2)$?

A 2

B 4

C 7

D 10

E 12

F 14

Solution **Answer E**

Composing the function with itself gives

$$\begin{aligned} f(f(x)) &= a(ax + b) + b \\ &= a^2x + b(a + 1) \end{aligned}$$

Since this equals $4x + 3$ for every real x, the corresponding coefficients must be equal. Hence

$$a^2 = 4 \quad \text{and} \quad b(a + 1) = 3$$

Since $a^2 = 4$ there are two possible values of a, $a = 2$ or $a = -2$
If $a = 2$, then $3b = 3$, so $b = 1$ and

$$f(2) = 2(2) + 1 = 5$$

If $a = -2$, then $-b = 3$, so $b = -3$ and

$$f(2) = -2(2) - 3 = -7$$

The greatest possible value is 5 and the least is -7, so their difference is

$$5 - (-7) = 12$$

Therefore the correct option is **E**.

Algebra

13. The equation $x^2 - 7x + 1 = 0$ has two positive solutions, p and q.

Find the value of

$$\frac{p-1}{p+1} \times \frac{q-1}{q+1}$$

A $-\frac{2}{3}$

B $-\frac{5}{8}$

C $-\frac{5}{9}$

D $\frac{5}{9}$

E $\frac{5}{8}$

F $\frac{2}{3}$

Solution **Answer C**

For a quadratic $ax^2 + bx + c = 0$, the sum of the roots is $-b/a$ and their product is c/a. Therefore

$$p + q = 7 \qquad \text{and} \qquad pq = 1$$

Combine the two fractions before substituting these values:

$$\frac{p-1}{p+1} \times \frac{q-1}{q+1} = \frac{(p-1)(q-1)}{(p+1)(q+1)} = \frac{pq - (p+q) + 1}{pq + (p+q) + 1}$$

Hence the required value is

$$\frac{1 - 7 + 1}{1 + 7 + 1} = -\frac{5}{9}$$

Thus the correct option is **C**.

Alternative Solution:

The quadratic formula can instead be used to find the two roots explicitly. Since

$$\begin{aligned} x &= \frac{7 \pm \sqrt{49 - 4}}{2} \\ &= \frac{7 \pm \sqrt{45}}{2} \\ &= \frac{7 \pm 3\sqrt{5}}{2} \end{aligned}$$

we may take

$$p = \frac{7 + 3\sqrt{5}}{2} \qquad \text{and} \qquad q = \frac{7 - 3\sqrt{5}}{2}$$

For each factor, subtracting or adding 1 gives expressions with the same denominator of 2, which therefore cancels within the fraction.

Hence

$$\begin{aligned}\frac{p-1}{p+1}\times\frac{q-1}{q+1} &= \frac{5+3\sqrt{5}}{9+3\sqrt{5}}\times\frac{5-3\sqrt{5}}{9-3\sqrt{5}}\\ &= \frac{(5+3\sqrt{5})(5-3\sqrt{5})}{(9+3\sqrt{5})(9-3\sqrt{5})}\\ &= \frac{25-45}{81-45}\\ &= -\frac{20}{36}\\ &= -\frac{5}{9}\end{aligned}$$

Again obtaining option **C**.

This route works, but the surd manipulation is longer and may waste time; using the sum and product of the roots is much more efficient.

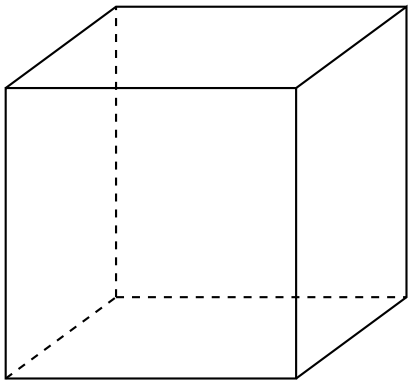
Probability & Counting

14. A cube has edge length 1.

Three different vertices are chosen at random.

Every choice of three vertices is equally likely.

What is the probability that the three chosen vertices form a right-angled triangle?



A $\frac{3}{7}$

B $\frac{1}{2}$

C $\frac{4}{7}$

D $\frac{3}{4}$

E $\frac{6}{7}$

F 1

Solution **Answer E**

The distance between two vertices of a cube is determined by how they are joined:

- vertices joined by an edge are distance 1
- vertices joined by a face diagonal are distance $\sqrt{2}$
- vertices joined by a space diagonal are distance $\sqrt{3}$

So every triangle formed by three vertices must have side lengths chosen from 1, $\sqrt{2}$ and $\sqrt{3}$.

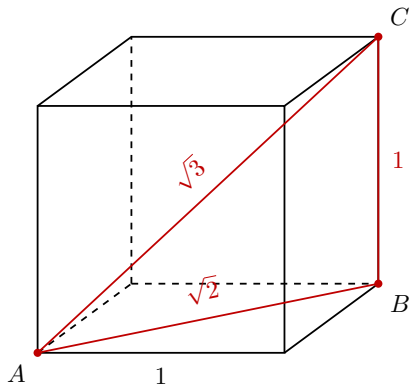
If a triangle contains a space diagonal, then its side lengths are

$$1,\ \sqrt{2},\ \sqrt{3}$$

and this is a right-angled triangle because

$$1^2 + (\sqrt{2})^2 = (\sqrt{3})^2$$

as shown in the following diagram:

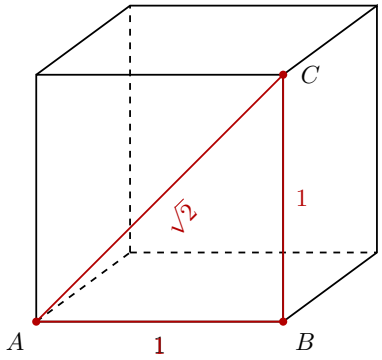


If there is no space diagonal but the triangle contains an edge, then the third side must be a face diagonal, so its side lengths are

$$1,\ 1,\ \sqrt{2}$$

which is again right-angled because

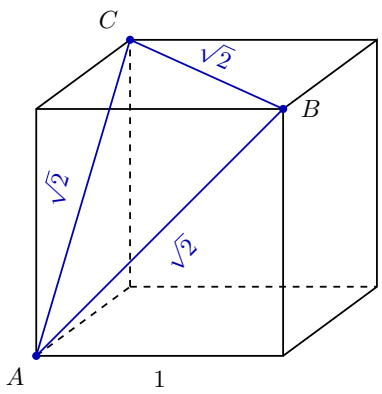
$$1^2 + 1^2 = (\sqrt{2})^2$$



Hence the only triangles which are *not* right-angled are those with no edge and no space diagonal, so all three sides must be face diagonals. Such a triangle has side lengths

$$\sqrt{2},\ \sqrt{2},\ \sqrt{2}$$

so it is equilateral, not right-angled:



We now count these non-right-angled triangles.

Fix the upper and lower faces of the cube. For any chosen vertex A, there are exactly two vertices on the opposite horizontal face which are a face diagonal away from A, and those two vertices are themselves a face diagonal apart. So each choice of A determines exactly one equilateral triangle of this type.

There are 8 choices for A, and each equilateral triangle is counted exactly once in this way, since it has a unique vertex on one horizontal face and the other two vertices on the opposite horizontal face. So there are

$$8$$

non-right-angled triangles.

The total number of ways to choose three vertices from the 8 vertices of the cube is

$$\binom{8}{3} = 56$$

Therefore the probability of obtaining a right-angled triangle is

$$1 - \frac{8}{56} = 1 - \frac{1}{7} = \frac{6}{7}$$

Therefore the correct option is **E**.

Probability & Counting

15. A box contains red counters and blue counters. The number of red counters exceeds the number of blue counters by an integer d.

Two counters are selected at random, one after the other, without replacement. The probability that the counters have the same colour is equal to the probability that they have different colours.

In terms of d, how many counters are in the box?

A $\frac{d^2}{2}$

B $d(d-1)$

C d^2

D $d(d+1)$

E $2d^2$

Solution **Answer C**

Let the numbers of red and blue counters be r and b, respectively, and let the total number be N. Then

$$r = b + d \qquad \text{and} \qquad N = r + b = 2b + d$$

The probability of selecting two counters of the same colour is

$$\frac{r(r-1) + b(b-1)}{N(N-1)}$$

For different colours, the red counter can be selected first or the blue counter can be selected first. These two possible orders give

$$\frac{rb + br}{N(N-1)} = \frac{2rb}{N(N-1)}$$

Equating the probabilities gives:

$$\frac{r(r-1) + b(b-1)}{N(N-1)} = \frac{2rb}{N(N-1)}$$

and cancelling the non-zero common denominator $N(N-1)$

$$r(r-1) + b(b-1) = 2rb$$

Now substitute $r = b + d$:

$$(b+d)(b+d-1) + b(b-1) = 2b(b+d)$$

Expanding and simplifying gives

$$d^2 - 2b - d = 0$$

Therefore

$$2b + d = d^2$$

But $2b + d = N$, so the box contains d^2 counters. The correct option is **C**.

Polynomials

16. Consider the curve C with equation

$$y = x^3 - 3x^2 + 1$$

Find the complete set of values of the constant c for which there are exactly three distinct lines passing through the point $(0, c)$ that are tangent to C.

A $-3 < c < 1$

B $-3 \leq c \leq 1$

C $1 < c < 2$

D $1 \leq c \leq 2$

E $c < -3$ or $c > 1$

F $c < 1$ or $c > 2$

Solution **Answer C**

Differentiating the curve gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 6x$$

Suppose the tangent touches the curve at $x = t$. Its gradient is $3t^2 - 6t$, and because it passes through $(0, c)$, its equation is

$$y = (3t^2 - 6t)x + c$$

At the point of tangency, both this line and the curve have the same value when $x = t$. Therefore

$$t^3 - 3t^2 + 1 = (3t^2 - 6t)t + c$$

so

$$c = -2t^3 + 3t^2 + 1$$

Thus the required number of tangents is determined by the number of distinct real solutions of $g(t) = c$, where

$$g(t) = -2t^3 + 3t^2 + 1$$

Distinct solutions give distinct tangent lines.

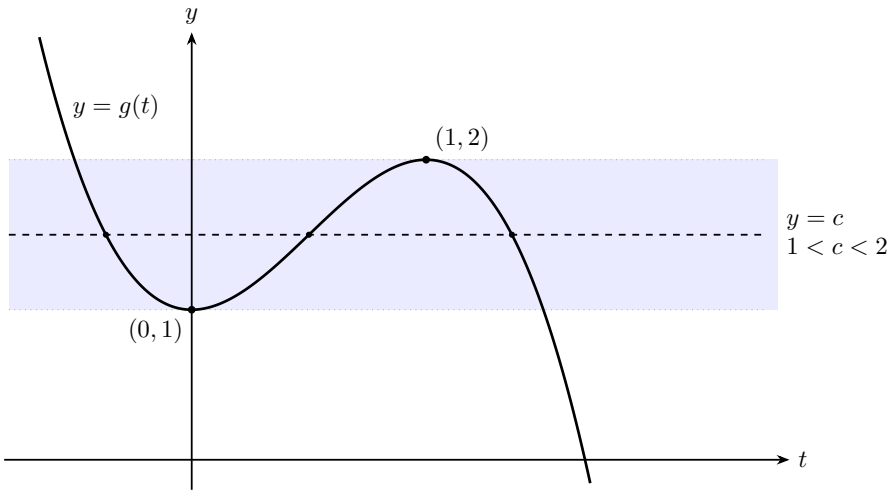
To find when $g(t) = c$ has three distinct solutions, differentiate:

$$g'(t) = -6t^2 + 6t = 6t(1 - t)$$

The stationary points occur at $t = 0$ and $t = 1$, with

$$g(0) = 1 \qquad \text{and} \qquad g(1) = 2$$

Since $g(t)$ has the shape of a cubic with negative x^3 coefficient it follows that $(0, 1)$ is a local minimum and $(1, 2)$ is a local maximum.



A horizontal line $y = c$ therefore crosses the graph of $y = g(t)$ three times precisely when it lies strictly between these stationary values. At $c = 1$ or $c = 2$, one intersection is repeated, so there are only two distinct points of tangency.

Hence the complete range is

$$1 < c < 2$$

the correct option is **C**.

Trigonometry

17. Find the complete set of real values of k for which the equation

$$2\cos^2 x - k\sin x + k = 0$$

has exactly one solution in the range $0 \leq x < 2\pi$.

A $k < -4$ or $k > 0$

B $k \leq -4$ or $k > 0$

C $k < -4$ or $k \geq 0$

D $k \leq -4$ or $k \geq 0$

E $-4 < k < 0$

F $-4 \leq k \leq 0$

Solution **Answer B**

Let $s = \sin x$. Using $\cos^2 x = 1 - \sin^2 x$, the equation becomes

$$2s^2 + ks - k - 2 = 0$$

To factorise with the parameter k, group the terms to produce the common factor $s - 1$:

$$(2s^2 - 2) + (ks - k) = 2(s-1)(s+1) + k(s-1)$$

Hence

$$(s-1)(2s + k + 2) = 0$$

Therefore

$$\sin x = 1 \quad \text{or} \quad \sin x = -\frac{k+2}{2}$$

Alternatively to factorising, using the quadratic formula:

$$s = \frac{-k \pm \sqrt{k^2 + 8k + 16}}{4} = \frac{-k \pm (k+4)}{4}$$

which gives the same two roots

$$\sin x = 1 \quad \text{or} \quad \sin x = -\frac{k+2}{2}$$

The first equation always gives exactly one solution in the given range, namely

$$x = \frac{\pi}{2}$$

Therefore, for the original equation to have exactly one solution, the second equation

$$\sin x = -\frac{k+2}{2}$$

must either have no solutions, or give no additional solution

Since

$$-1 \leq \sin x \leq 1$$

the second equation has no solutions when

$$-\frac{k+2}{2} > 1 \qquad \text{or} \qquad -\frac{k+2}{2} < -1$$

It also gives no additional solution when

$$-\frac{k+2}{2} = 1$$

since this simply gives the same equation $\sin x = 1$

However,

$$-\frac{k+2}{2} = -1$$

does not qualify, because $\sin x = -1$ gives the additional solution

$$x = \frac{3\pi}{2}$$

The required cases are therefore

$$-\frac{k+2}{2} > 1, \qquad -\frac{k+2}{2} < -1, \qquad \text{or} \qquad -\frac{k+2}{2} = 1$$

These give, respectively,

$$k < -4, \qquad k > 0, \qquad \text{or} \qquad k = -4$$

Combining them, the complete set is

$$k \leq -4 \quad \text{or} \quad k > 0$$

so the correct option is **B**.

Alternative Solution: (Testing the options)

Every option either includes or excludes each of the boundary values $k = -4$ and $k = 0$, so test these two values first.

First take $k = -4$. The equation becomes

$$2\cos^2 x + 4\sin x - 4 = 0$$

Using $\cos^2 x = 1 - \sin^2 x$ gives

$$2\sin^2 x - 4\sin x + 2 = 0$$

and hence

$$2(\sin x - 1)^2 = 0$$

Thus

$$\sin x = 1$$

which has exactly one solution in the required range:

$$x = \frac{\pi}{2}$$

Therefore $k = -4$ must be included. This eliminates options **A**, **C** and **E**.

Now take $k = 0$. The equation becomes

$$2\cos^2 x = 0$$

so

$$\cos x = 0$$

This has two solutions in the required range:

$$x = \frac{\pi}{2},\ \frac{3\pi}{2}$$

Therefore $k = 0$ must not be included. Of the remaining options **B**, **D** and **F**, this eliminates **D** and **F**.

The only remaining option is **B**.

Polynomials

18. Let $f(x) = x^4 + ax^2 + b$, where a and b are real constants.

The minimum value of $f(x)$ is -4, and the equation $f(x) = 0$ has exactly three distinct real solutions.

What is the value of a and the value of b?

A $a = -4$, $b = -4$

B $a = -4$, $b = 0$

C $a = -2$, $b = 0$

D $a = 0$, $b = -4$

E $a = 4$, $b = -4$

F $a = 4$, $b = 0$

Solution **Answer B**

Let $u = x^2$, so $u \geq 0$. The equation $f(x) = 0$ becomes

$$u^2 + au + b = 0$$

A positive solution $u = r$ produces the two distinct real solutions $x = \sqrt{r}$ and $x = -\sqrt{r}$, whereas $u = 0$ produces only the single solution $x = 0$.

Since there are exactly three distinct real solutions in x, the two relevant solutions in u must therefore be $u = 0$ and $u = r$ for some $r > 0$.

Substituting $u = 0$ gives $b = 0$.

The equation then factorises as

$$u^2 + au = u(u + a) = 0$$

so its other solution is $u = -a$. As this solution is positive, $a < 0$.

Now complete the square:

$$f(x) = u^2 + au$$

$$= \left(u + \frac{a}{2}\right)^2 - \frac{a^2}{4}$$

Because $a < 0$, the value $u = -\dfrac{a}{2}$ is positive and hence is attainable by an appropriate value of x.

Therefore, from the completed square form, the minimum value is $-\dfrac{a^2}{4}$. Thus

$$-\frac{a^2}{4} = -4$$

so $a^2 = 16$. The condition $a < 0$ selects $a = -4$, and $b = 0$. Hence the correct option is **B**.

Alternative Solution:

$f(x)$ depends only on x^2. Thus the function $f(x)$ is an even function, since $f(-x) = f(x)$.

Recall that even functions are exactly those functions with graphs for which the y-axis is a line of symmetry.

Consequently, every non-zero root occurs with its reflection in the y-axis: if $x = c$ is a root, then $x = -c$ is also a root.

An odd number of distinct roots is therefore possible only if $x = 0$ is also a root.

Hence $f(0) = 0$. But $f(0) = b$ so $b = 0$.

Since the minimum is negative we must have $a < 0$, as otherwise:

$$f(x) = x^4 + ax^2 \geq 0$$

for all x. Completing the square gives

$$f(x) = \left(x^2 + \frac{a}{2}\right)^2 - \frac{a^2}{4}$$

The squared term can equal zero because $-a/2 > 0$.

The minimum is therefore $-a^2/4 = -4$, giving $a = -4$.

Together with $b = 0$, this again gives **B**.

Functions & Graphs

19. For a positive real constant k, the equation

$$|x^2 - 4x + 3| = kx$$

has exactly three distinct real solutions.

Find the complete set of possible values of k.

A There are no such positive values of k.

B $0 < k < 4 - 2\sqrt{3}$

C $k = 4 - 2\sqrt{3}$

D $k = 4 + 2\sqrt{3}$

E $k = 4 - 2\sqrt{3}$ or $k = 4 + 2\sqrt{3}$

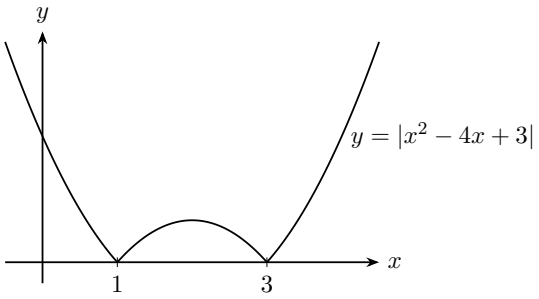
Solution **Answer C**

Write

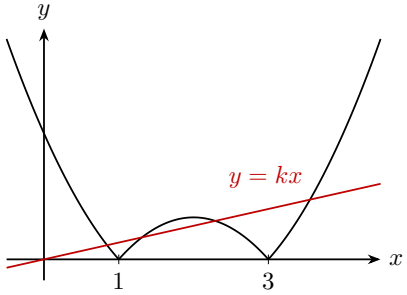
$$q(x) = x^2 - 4x + 3 = (x - 1)(x - 3)$$

The quadratic $q(x)$ is negative between its roots 1 and 3. The absolute value reflects this part above the x-axis, so

$$|x^2 - 4x + 3| = \begin{cases} x^2 - 4x + 3, & x \leq 1 \text{ or } x \geq 3, \\ -x^2 + 4x - 3, & 1 \leq x \leq 3. \end{cases}$$



For $k > 0$, the graph shows that the line $y = kx$ meets each of the two upward-facing branches exactly once. Thus these branches always contribute two solutions:



For there to be exactly three distinct solutions altogether, the line must therefore meet the central arch exactly once. This occurs when $y = kx$ is tangent to the central arch.

We therefore need exactly one further solution on the central arch. There,

$$kx = -x^2 + 4x - 3$$

so

$$x^2 - (4 - k)x + 3 = 0$$

The line must be tangent to the arch, giving a repeated root. Equivalently, the discriminant must be zero:

$$(4 - k)^2 - 12 = 0$$

Hence

$$4 - k = \pm 2\sqrt{3}$$

and the two candidates are

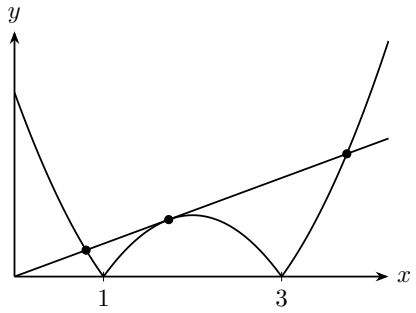
$$k = 4 - 2\sqrt{3} \quad \text{or} \quad k = 4 + 2\sqrt{3}$$

Care is needed because the central formula is valid only for $1 \leq x \leq 3$.

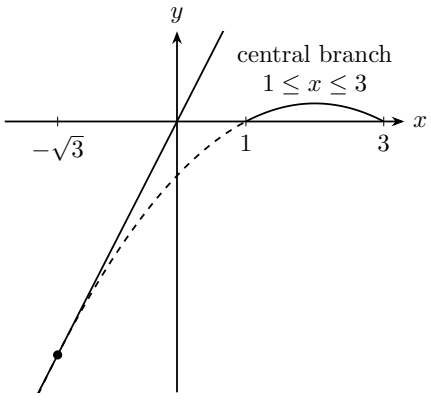
The repeated root of a quadratic with zero discriminant is given by $-\dfrac{b}{2a}$. In this case:

$$x = \frac{4 - k}{2}$$

For $k = 4 - 2\sqrt{3}$, this gives $x = \sqrt{3}$, which lies between 1 and 3. This is the required third intersection:



For $k = 4 + 2\sqrt{3}$, the repeated root is $x = -\sqrt{3}$. This is a tangency to the extension of the central parabola, not to the part used in the absolute-value graph. There is therefore no central intersection for this value of k:



Consequently, the complete set of possible values is $k = 4 - 2\sqrt{3}$, so the correct option is **C**.

Geometry

20. A circle passes through the vertices of a right-angled triangle ABC, where angle $ABC = 90°$, $AB = 12$ cm, and $BC = 16$ cm.

A line is drawn through A parallel to BC.

This line intersects the tangent to the circle at B at the point P.

What is the length of AP?

A 7.2 cm

B 8 cm

C 9 cm

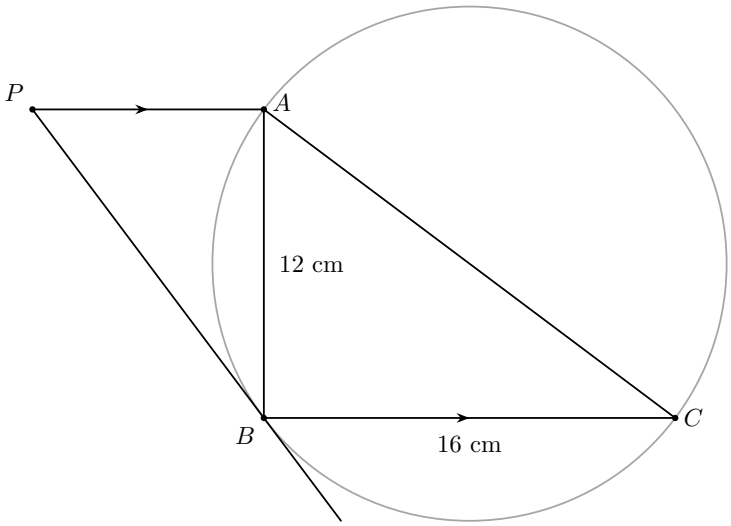
D 9.6 cm

E 10 cm

F 16 cm

Solution **Answer C**

A diagram with the two constructed lines is shown below:



Since PB is tangent to the circle at B and BA is a chord, the alternate segment theorem gives

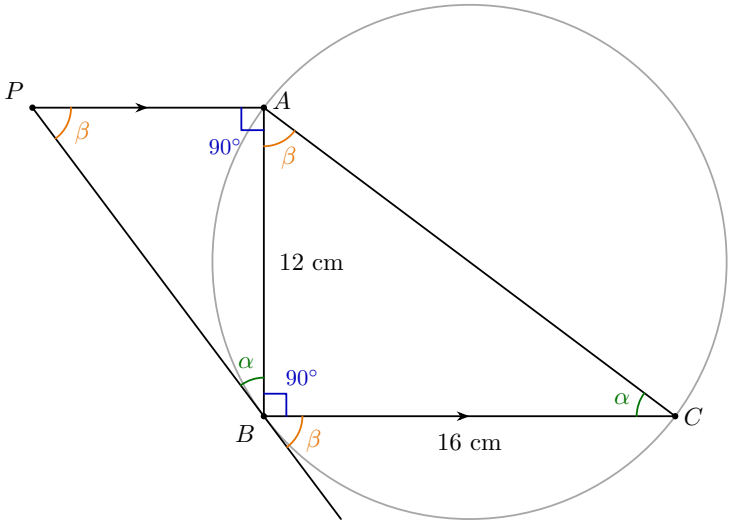
$$\angle PBA = \angle BCA$$

Also, AP is parallel to BC. Since AB is perpendicular to BC, it is also perpendicular to AP, so

$$\angle PAB = \angle ABC = 90°$$

The remaining angles are therefore equal as well, because the angles in each triangle sum to

180°. The corresponding angles are marked with the same colour and symbol in the diagram below.



Thus triangles PBA and ABC are similar. In triangle PBA, the vertices P, B, A correspond respectively to A, C, B in triangle ABC. Therefore AB in triangle PBA corresponds to BC in triangle ABC, while PA corresponds to AB in triangle ABC.

The scale factor from triangle ABC to triangle PBA is consequently

$$\frac{12}{16} = \frac{3}{4}$$

Since PA corresponds to the 12 cm side AB in triangle ABC,

$$PA = \frac{3}{4} \times 12 = 9 \text{ cm}$$

Hence the correct option is **C**.